MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **Periodic Test Paper 11 (Cumulative)** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 5: Arithmetic Progressions** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

General Instructions:

- This question paper consists of **12 questions** divided into **Five Sections**: A, B, C, D, and E.
- **Section A** comprises **5 questions** of **1 mark** each (Q1–Q4 MCQs, Q5 Assertion-Reason).
- **Section B** comprises **2 questions** (Q6–Q7) of **2 marks** each.
- **Section C** comprises **2 questions** (Q8–Q9) of **3 marks** each.
- **Section D** comprises **2 questions** (Q10–Q11) of **5 marks** each.
- **Section E** comprises **1 Applied Case Study** (Q12) of **5 marks** with internal choice in sub-part (iii).
- All questions are compulsory. Use of calculators is strictly prohibited.

SECTION A: Multiple Choice & Assertion-Reason Questions (5 × 1 = 5 Marks)

Q1. How many terms of the AP $9, 17, 25, \ldots$ must be taken to give a sum of 636? [1]

(A) 11 (B) 12 (C) 13 (D) 14

Q2. The first and last terms of an AP are 17 and 350 respectively. If the common difference is 9, how many terms are there in the AP? [1]

(A) 36 (B) 37 (C) 38 (D) 39

Q3. In an AP, if $a_n = 4n - 5$, then the sum of its first 25 terms is: [1]

(A) 1175 (B) 1150 (C) 1200 (D) 1225

Q4. If $p - 1, p + 3, 3p - 1$ are in arithmetic progression, then the value of p is: [1]

(A) 4 (B) -4 (C) 2 (D) -2

Q5. **Assertion (A):** The sequence $\sqrt{3}, \sqrt{12}, \sqrt{27}, \sqrt{48}, \ldots$ is an arithmetic progression.
Reason (R): Simplifying the surds yields $\sqrt{3}, 2\sqrt{3}, 3\sqrt{3}, 4\sqrt{3}, \ldots$, which has a constant common difference $d = \sqrt{3}$. [1]

(A) Both (A) and (R) are true, and (R) is correct explanation.
(B) Both (A) and (R) are true, but (R) is not correct explanation.
(C) Assertion (A) is true, but Reason (R) is false.
(D) Assertion (A) is false, but Reason (R) is true.

SECTION B: Very Short Answer Questions (2 × 2 = 4 Marks)

Q6. Find the sum of the first 15 positive multiples of 8. [2]

Q7. If $a_n = 9 - 5n$ represents the nth term of an AP, find the sum of its first 15 terms. [2]

SECTION C: Short Answer Questions (2 × 3 = 6 Marks)

Q8. Find three numbers in AP whose sum is 21 and the product of the first and the third number exceeds the second number by 6. [3]

Q9. Find the 4th term from the end of the arithmetic progression $-11, -8, -5, \ldots, 49$. [3]

SECTION D: Long Answer / HOTS Questions (2 × 5 = 10 Marks)

Q10. A manufacturer of laptop batteries produced 600 units in the third year and 700 units in the seventh year. Assuming that the production increases uniformly by a constant number every year, calculate:

(a) The production in the first year.

(b) The production in the 10th year.

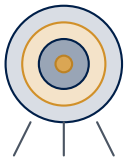
(c) The total cumulative production in the first 7 years.

[5]

Q11. If the sum of first p terms of an AP is equal to q and the sum of first q terms is equal to p, prove that the sum of its first $(p+q)$ terms is $-(p+q)$. [5]

SECTION E: Applied Case Study (1 × 5 = 5 Marks)

Q12. CASE STUDY: State Championship Archery Target Practice



An archer preparing for the National Youth Archery Championship designs a progressive daily target shooting training schedule to build shoulder and core endurance.
On Day 1 of the training camp, she shoots 30 arrows. To prevent overtraining while progressively overloading muscle capacity, she shoots 5 additional arrows each succeeding day throughout the 21-day training camp.

Based on the above sports conditioning regimen, answer the following questions:

(i) Find the number of arrows shot by the archer on Day 6 of the camp (a_6). [1]

(ii) On which day of the training camp will she shoot exactly 100 arrows? [1]

(iii) Calculate the total number of arrows shot by the archer during the entire 21-day training camp (S_{21}). [3]

OR

(iv) Find the ratio of the number of arrows shot on Day 21 to the number of arrows shot on Day 1. [3]

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★